

Evaluation Statistics Report
Input CSV: /Users/julian/CodingProjects/Unit3/QuarticSpeedupK-XOR/logs/summary_all_instances.csv
Generated: 2026-01-20 12:27:31.089454

Evaluation Statistics Report
Input CSV: /Users/julian/CodingProjects/Unit3/QuarticSpeedupK-XOR/logs/summary_all_instances.csv

Total rows: 846
Successful: 846 (success_rate=1.000)

Note: rho=0 rows detected. If those instances have no ground truth, eval_* metrics will be NaN. This report excludes NaNs from metric statistics, so rho=0 will not distort evaluation aggregates.

Metrics summarized (on successful rows, excluding NaNs):
eval_advantage, eval_correlation, eval_hamming_frac, eval_bit_accuracy_est

=== overall ===

All instances together. Includes failures for success statistics; metric distributions are computed only over successful rows where the metric exists.

(notebook) Compact overall summary:

	metric	n_total	n_success	success_rate	n_metric
c	mean	median	p10	p90	
eval_advantage		846	846	1.0	84
6	0.542322	0.486905	0.221227	1.0	

-- eval_advantage --

n_total	n_success	success_rate	n_metric	mean	median
846	846	1.0	846	0.542322	0.486905
0.285141	0.798895	0.513754	0.288559	-0.264957	1.0

-- eval_correlation --

n_total	n_success	success_rate	n_metric	mean	median
846	846	1.0	705	0.802145	1.0
0.7	1.0	0.3	0.301188	0.0	1.0

-- eval_hamming_frac --

n_total	n_success	success_rate	n_metric	mean	median
846	846	1.0	705	0.098927	0.0
0.0	0.0	0.15	0.15	0.150594	0.0

-- eval_bit_accuracy_est --

n_total	n_success	success_rate	n_metric	mean	median
846	846	1.0	846	0.771161	0.74
3453	0.64257	0.899447	0.256877	0.14428	0.367521

=== by_rho ===

Grouped by rho (noise level). For rho=0, ground truth may be absent, so eval_* metrics can be NaN; those rows do not contribute to metric aggregates.

Groups: 6 (metrics computed over successful rows with non-NaN metric values)

-- eval_advantage --

rho	n_total	n_success	success_rate	n_metric	mean
-----	---------	-----------	--------------	----------	------

	median	p25	p75	iqr	std	min
0.0	141	141	141	1.0	141	0.241732
0.244681	0.213115	0.289720	0.076605	0.089306	-0.264957	0.413333
0.2	141	141	141	1.0	141	0.265388
0.271111	0.233645	0.301370	0.067725	0.060623	0.106383	0.397959
0.4	141	141	141	1.0	141	0.371293
0.378378	0.305085	0.447964	0.142879	0.093288	0.151515	0.616766
0.6	141	141	141	1.0	141	0.578023
0.582677	0.530864	0.621622	0.090757	0.069897	0.395745	0.764706
0.8	141	141	141	1.0	141	0.797498
0.799043	0.777778	0.820628	0.042850	0.042488	0.696000	0.906433
1.0	141	141	141	1.0	141	1.000000
1.000000	1.000000	1.000000	0.000000	0.000000	1.000000	1.000000

-- eval_correlation --

rho	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min
0.0	141	141	1.0	0	NaN
NaN	NaN	NaN	NaN	NaN	NaN
0.2	141	141	1.0	141	0.335318
0.294118	0.142857	0.500000	0.357143	0.236477	0.000
0.4	141	141	1.0	141	0.721713
0.785714	0.636364	0.894737	0.258373	0.256420	0.000
0.6	141	141	1.0	141	0.954583
1.000000	0.904762	1.000000	0.095238	0.080468	0.500
0.8	141	141	1.0	141	0.999113
1.000000	1.000000	1.000000	0.000000	0.010527	0.875
1.0	141	141	1.0	141	1.000000
1.000000	1.000000	1.000000	0.000000	0.000000	1.000

-- eval_hamming_frac --

rho	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min
0.0	141	141	1.0	0	NaN
NaN	NaN	NaN	NaN	NaN	NaN
0.2	141	141	1.0	141	0.332341
0.352941	0.250000	0.428571	0.178571	0.118238	0.0
0.4	141	141	1.0	141	0.139143
0.107143	0.052632	0.181818	0.129187	0.128210	0.0
0.6	141	141	1.0	141	0.022709
0.000000	0.000000	0.047619	0.047619	0.040234	0.0
0.8	141	141	1.0	141	0.000443
0.000000	0.000000	0.000000	0.000000	0.005263	0.0
1.0	141	141	1.0	141	0.000000
0.000000	0.000000	0.000000	0.000000	0.000000	0.0

-- eval_bit_accuracy_est --

rho	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min
0.0	141	141	1.0	141	0.620866
0.622340	0.606557	0.644860	0.038302	0.044653	0.367521
0.2	141	141	1.0	141	0.632694
0.635556	0.616822	0.650685	0.033863	0.030312	0.553191
0.4	141	141	1.0	141	0.685646

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0.689189 0.652542 0.723982 0.071440 0.046644 0.575758 0.
808383
0.6      141      141      1.0      141 0.789012
0.791339 0.765432 0.810811 0.045379 0.034948 0.697872 0.
882353
0.8      141      141      1.0      141 0.898749
0.899522 0.888889 0.910314 0.021425 0.021244 0.848000 0.
953216
1.0      141      141      1.0      141 1.000000
1.000000 1.000000 1.000000 0.000000 0.000000 1.000000 1.
000000

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=== by_rho === (compact)
eval_advantage (top 8 groups by n_total):
rho  n_total  n_success  success_rate  n_metric  mean
median
0.0      141      141      1.0      141 0.241732
0.244681
0.2      141      141      1.0      141 0.265388
0.271111
0.4      141      141      1.0      141 0.371293
0.378378
0.6      141      141      1.0      141 0.578023
0.582677
0.8      141      141      1.0      141 0.797498
0.799043
1.0      141      141      1.0      141 1.000000
1.000000

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=== by_kappa ===
Grouped by kappa (signal strength / plantedness parameter
).
Groups: 4 (metrics computed over successful rows with non
-NaN metric values)
-- eval_advantage --
kappa  n_total  n_success  success_rate  n_metric  me
an  median  p25  p75  iqr  std  min
max
0.625      18      18      1.0      18 0.5122
53 0.538153 0.166667 0.749221 0.582554 0.333251 0.120332
1.0
0.750      132      132      1.0      132 0.5337
66 0.488271 0.259292 0.812308 0.553016 0.297600 0.052023
1.0
0.875      294      294      1.0      294 0.5327
45 0.477589 0.269952 0.796610 0.526658 0.295191 -0.250000
1.0
1.000      402      402      1.0      402 0.5534
83 0.497126 0.301108 0.802554 0.501446 0.279097 -0.264957
1.0

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-- eval_correlation --
kappa  n_total  n_success  success_rate  n_metric  me
an  median  p25  p75  iqr  std  min  max
0.625      18      18      1.0      15 0.8065
90 1.0 0.764706 1.0 0.235294 0.340576 0.000000 1.0
0.750      132      132      1.0      110 0.8292
64 1.0 0.780702 1.0 0.219298 0.274776 0.047619 1.0
0.875      294      294      1.0      245 0.8022
18 1.0 0.727273 1.0 0.272727 0.297190 0.000000 1.0
1.000      402      402      1.0      335 0.7929
88 1.0 0.644269 1.0 0.355731 0.311214 0.000000 1.0

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-- eval_hamming_frac --

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kappa	n_total	n_success	success_rate	n_metric	me		
an	median	p25	p75	iqr	std	min	max
0.625	18	18	1.0	15	0.0967		
05	0.0	0.0	0.117647	0.117647	0.170288	0.0	0.50000
0.750	132	132	1.0	110	0.0853		
68	0.0	0.0	0.109649	0.109649	0.137388	0.0	0.47619
0.875	294	294	1.0	245	0.0988		
91	0.0	0.0	0.136364	0.136364	0.148595	0.0	0.50000
1.000	402	402	1.0	335	0.1035		
06	0.0	0.0	0.177866	0.177866	0.155607	0.0	0.50000

-- eval_bit_accuracy_est --

kappa	n_total	n_success	success_rate	n_metric	me		
an	median	p25	p75	iqr	std	min	max
0.625	18	18	1.0	18	0.7561		
27	0.769076	0.583333	0.874610	0.291277	0.166625	0.560166	
1.0							
0.750	132	132	1.0	132	0.7668		
83	0.744135	0.629646	0.906154	0.276508	0.148800	0.526012	
1.0							
0.875	294	294	1.0	294	0.7663		
72	0.738795	0.634976	0.898305	0.263329	0.147595	0.375000	
1.0							
1.000	402	402	1.0	402	0.7767		
41	0.748563	0.650554	0.901277	0.250723	0.139548	0.367521	
1.0							

=== by_kappa === (compact)

eval_advantage (top 8 groups by n_total):

kappa	n_total	n_success	success_rate	n_metric	me
an	median				
1.000	402	402	1.0	402	0.5534
83	0.497126				
0.875	294	294	1.0	294	0.5327
45	0.477589				
0.750	132	132	1.0	132	0.5337
66	0.488271				
0.625	18	18	1.0	18	0.5122
53	0.538153				

=== by_rho_kappa ===

2D grouping by (rho, kappa) to expose interaction between noise and planted strength.

Groups: 24 (metrics computed over successful rows with no n-NaN metric values)

-- eval_advantage --

rho	kappa	n_total	n_success	success_rate	n_metric	me
mean	median	p25	p75	iqr	std	
min	max					
0.0	0.625	3	3	1.0	3	0
.137365	0.135135	0.127734	0.145881	0.018147	0.018250	0.1
20332	0.156627					
0.0	0.750	22	22	1.0	22	0
.212379	0.214621	0.199519	0.251267	0.051748	0.053179	0.0
52023	0.287234					
0.0	0.875	49	49	1.0	49	0
.226805	0.237569	0.203883	0.278912	0.075028	0.091798	-0.2
50000	0.352941					
0.0	1.000	67	67	1.0	67	0
.266960	0.273973	0.229842	0.318885	0.089042	0.091025	-0.2
64957	0.413333					
0.2	0.625	3	3	1.0	3	0
.213504	0.180180	0.156355	0.253991	0.097635	0.101811	0.1

32530	0.327801						
0.2	0.750	22	22	1.0		22	0
.236119	0.250616	0.220309	0.261183	0.040873	0.047105	0.1	
06383	0.302326						
0.2	0.875	49	49	1.0		49	0
.250568	0.254902	0.193717	0.297710	0.103993	0.068451	0.1	
16466	0.397959						
0.2	1.000	67	67	1.0		67	0
.288160	0.289855	0.262757	0.319091	0.056334	0.047652	0.1	
75510	0.385714						
0.4	0.625	3	3	1.0		3	0
.306116	0.286307	0.224235	0.378093	0.153859	0.154812	0.1	
62162	0.469880						
0.4	0.750	22	22	1.0		22	0
.373039	0.365289	0.317151	0.430060	0.112908	0.083792	0.1	
96653	0.492308						
0.4	0.875	49	49	1.0		49	0
.352500	0.376518	0.288136	0.435407	0.147271	0.094919	0.1	
68000	0.485944						
0.4	1.000	67	67	1.0		67	0
.387382	0.385542	0.325216	0.457571	0.132355	0.090711	0.1	
51515	0.616766						
0.6	0.625	3	3	1.0		3	0
.626500	0.621622	0.614024	0.636537	0.022513	0.022906	0.6	
06426	0.651452						
0.6	0.750	22	22	1.0		22	0
.575152	0.587488	0.539069	0.615981	0.076912	0.061214	0.4	
44444	0.672414						
0.6	0.875	49	49	1.0		49	0
.574217	0.578231	0.518987	0.611814	0.092827	0.070296	0.4	
05858	0.734043						
0.6	1.000	67	67	1.0		67	0
.579579	0.580357	0.539416	0.621044	0.081628	0.073747	0.3	
95745	0.764706						
0.8	0.625	3	3	1.0		3	0
.790034	0.759336	0.739106	0.825614	0.086508	0.090501	0.7	
18876	0.891892						
0.8	0.750	22	22	1.0		22	0
.805908	0.813505	0.786059	0.820800	0.034741	0.038382	0.7	
21311	0.875000						
0.8	0.875	49	49	1.0		49	0
.792380	0.796610	0.768421	0.813333	0.044912	0.037360	0.6	
99346	0.867925						
0.8	1.000	67	67	1.0		67	0
.798814	0.803922	0.777778	0.830952	0.053175	0.045275	0.6	
96000	0.906433						
1.0	0.625	3	3	1.0		3	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0	
00000	1.000000						
1.0	0.750	22	22	1.0		22	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0	
00000	1.000000						
1.0	0.875	49	49	1.0		49	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0	
00000	1.000000						
1.0	1.000	67	67	1.0		67	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0	
00000	1.000000						

-- eval_correlation --

rho	kappa	n_total	n_success	success_rate	n_metric
mean	median		p25	p75	std
min	max				
0.0	0.625	3	3	1.0	0

NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	22	22	1.0	0	
0.0	0.750					
NaN	NaN	NaN	NaN	NaN	NaN	
NaN	NaN	49	49	1.0	0	
0.0	0.875					
NaN	NaN	NaN	NaN	NaN	NaN	
NaN	NaN	67	67	1.0	0	
0.0	1.000					
NaN	NaN	NaN	NaN	NaN	NaN	
NaN	NaN	3	3	1.0	3	0
0.2	0.625					
.403050	0.444444	0.222222	0.604575	0.382353	0.384030	0.00
0000	0.764706					
0.2	0.750	22	22	1.0	22	0
.389952	0.333333	0.194853	0.570574	0.375721	0.242778	0.04
7619	0.857143					
0.2	0.875	49	49	1.0	49	0
.351634	0.333333	0.153846	0.500000	0.346154	0.242487	0.00
0000	0.913043					
0.2	1.000	67	67	1.0	67	0
.302413	0.259259	0.125217	0.470175	0.344958	0.223906	0.00
0000	1.000000					
0.4	0.625	3	3	1.0	3	0
.629902	0.764706	0.444853	0.882353	0.437500	0.452808	0.12
5000	1.000000					
0.4	0.750	22	22	1.0	22	0
.784099	0.838095	0.712500	0.905502	0.193002	0.209648	0.26
3158	1.000000					
0.4	0.875	49	49	1.0	49	0
.713149	0.769231	0.666667	0.888889	0.222222	0.262545	0.00
0000	1.000000					
0.4	1.000	67	67	1.0	67	0
.711603	0.777778	0.600000	0.911067	0.311067	0.259432	0.00
0000	1.000000					
0.6	0.625	3	3	1.0	3	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
0.6	0.750	22	22	1.0	22	0
.972266	1.000000	0.931818	1.000000	0.068182	0.046634	0.88
2353	1.000000					
0.6	0.875	49	49	1.0	49	0
.948861	1.000000	0.900000	1.000000	0.100000	0.069073	0.73
3333	1.000000					
0.6	1.000	67	67	1.0	67	0
.950927	1.000000	0.927249	1.000000	0.072751	0.096629	0.50
0000	1.000000					
0.8	0.625	3	3	1.0	3	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
0.8	0.750	22	22	1.0	22	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
0.8	0.875	49	49	1.0	49	0
.997449	1.000000	1.000000	1.000000	0.000000	0.017857	0.87
5000	1.000000					
0.8	1.000	67	67	1.0	67	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
1.0	0.625	3	3	1.0	3	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
1.0	0.750	22	22	1.0	22	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00

```

0000 1.000000
1.0 0.875 49 49 1.0 49 1
.000000 1.000000 1.000000 1.000000 0.000000 0.000000 1.00
0000 1.000000
1.0 1.000 67 67 1.0 67 1
.000000 1.000000 1.000000 1.000000 0.000000 0.000000 1.00
0000 1.000000

```

```

-- eval_hamming_frac --
rho kappa n_total n_success success_rate n_metric
mean median p25 p75 iqr std
min max
0.0 0.625 3 3 1.0 0
NaN NaN NaN NaN NaN NaN
NaN NaN
0.0 0.750 22 22 1.0 0
NaN NaN NaN NaN NaN NaN
NaN NaN
0.0 0.875 49 49 1.0 0
NaN NaN NaN NaN NaN NaN
NaN NaN
0.0 1.000 67 67 1.0 0
NaN NaN NaN NaN NaN NaN
NaN NaN
0.2 0.625 3 3 1.0 3 0
.298475 0.277778 0.197712 0.388889 0.191176 0.192015 0.11
7647 0.500000
0.2 0.750 22 22 1.0 22 0
.305024 0.333333 0.214713 0.402574 0.187861 0.121389 0.07
1429 0.476190
0.2 0.875 49 49 1.0 49 0
.324183 0.333333 0.250000 0.423077 0.173077 0.121244 0.04
3478 0.500000
0.2 1.000 67 67 1.0 67 0
.348794 0.370370 0.264912 0.437391 0.172479 0.111953 0.00
0000 0.500000
0.4 0.625 3 3 1.0 3 0
.185049 0.117647 0.058824 0.277574 0.218750 0.226404 0.00
0000 0.437500
0.4 0.750 22 22 1.0 22 0
.107950 0.080952 0.047249 0.143750 0.096501 0.104824 0.00
0000 0.368421
0.4 0.875 49 49 1.0 49 0
.143426 0.115385 0.055556 0.166667 0.111111 0.131272 0.00
0000 0.500000
0.4 1.000 67 67 1.0 67 0
.144199 0.111111 0.044466 0.200000 0.155534 0.129716 0.00
0000 0.500000
0.6 0.625 3 3 1.0 3 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
0.6 0.750 22 22 1.0 22 0
.013867 0.000000 0.000000 0.034091 0.034091 0.023317 0.00
0000 0.058824
0.6 0.875 49 49 1.0 49 0
.025570 0.000000 0.000000 0.050000 0.050000 0.034537 0.00
0000 0.133333
0.6 1.000 67 67 1.0 67 0
.024536 0.000000 0.000000 0.036376 0.036376 0.048315 0.00
0000 0.250000
0.8 0.625 3 3 1.0 3 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
0.8 0.750 22 22 1.0 22 0

```

```

.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
0.8 0.875 49 49 1.0 49 0
.001276 0.000000 0.000000 0.000000 0.000000 0.008929 0.00
0000 0.062500
0.8 1.000 67 67 1.0 67 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
1.0 0.625 3 3 1.0 3 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
1.0 0.750 22 22 1.0 22 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
1.0 0.875 49 49 1.0 49 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000
1.0 1.000 67 67 1.0 67 0
.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.00
0000 0.000000

```

-- eval_bit_accuracy_est --

	rho	kappa	n_total	n_success	success_rate	n_metric
	mean	median	p25	p75	iqr	std
	min	max				
0.0	0.625	3	3	1.0	3	0
.568682	0.567568	0.563867	0.572940	0.009074	0.009125	0.56
0166	0.578313					
0.0	0.750	22	22	1.0	22	0
.606190	0.607310	0.599760	0.625634	0.025874	0.026590	0.52
6012	0.643617					
0.0	0.875	49	49	1.0	49	0
.613402	0.618785	0.601942	0.639456	0.037514	0.045899	0.37
5000	0.676471					
0.0	1.000	67	67	1.0	67	0
.633480	0.636986	0.614921	0.659442	0.044521	0.045512	0.36
7521	0.706667					
0.2	0.625	3	3	1.0	3	0
.606752	0.590090	0.578178	0.626995	0.048818	0.050906	0.56
6265	0.663900					
0.2	0.750	22	22	1.0	22	0
.618059	0.625308	0.610155	0.630591	0.020437	0.023552	0.55
3191	0.651163					
0.2	0.875	49	49	1.0	49	0
.625284	0.627451	0.596859	0.648855	0.051996	0.034225	0.55
8233	0.698980					
0.2	1.000	67	67	1.0	67	0
.644080	0.644928	0.631379	0.659545	0.028167	0.023826	0.58
7755	0.692857					
0.4	0.625	3	3	1.0	3	0
.653058	0.643154	0.612117	0.689047	0.076929	0.077406	0.58
1081	0.734940					
0.4	0.750	22	22	1.0	22	0
.686519	0.682644	0.658576	0.715030	0.056454	0.041896	0.59
8326	0.746154					
0.4	0.875	49	49	1.0	49	0
.676250	0.688259	0.644068	0.717703	0.073636	0.047460	0.58
4000	0.742972					
0.4	1.000	67	67	1.0	67	0
.693691	0.692771	0.662608	0.728786	0.066178	0.045355	0.57
5758	0.808383					
0.6	0.625	3	3	1.0	3	0
.813250	0.810811	0.807012	0.818268	0.011257	0.011453	0.80
3213	0.825726					

0.6	0.750	22	22	1.0	22	0
.787576	0.793744	0.769534	0.807991	0.038456	0.030607	0.72
2222	0.836207					
0.6	0.875	49	49	1.0	49	0
.787108	0.789116	0.759494	0.805907	0.046414	0.035148	0.70
2929	0.867021					
0.6	1.000	67	67	1.0	67	0
.789790	0.790179	0.769708	0.810522	0.040814	0.036873	0.69
7872	0.882353					
0.8	0.625	3	3	1.0	3	0
.895017	0.879668	0.869553	0.912807	0.043254	0.045251	0.85
9438	0.945946					
0.8	0.750	22	22	1.0	22	0
.902954	0.906753	0.893029	0.910400	0.017370	0.019191	0.86
0656	0.937500					
0.8	0.875	49	49	1.0	49	0
.896190	0.898305	0.884211	0.906667	0.022456	0.018680	0.84
9673	0.933962					
0.8	1.000	67	67	1.0	67	0
.899407	0.901961	0.888889	0.915476	0.026587	0.022638	0.84
8000	0.953216					
1.0	0.625	3	3	1.0	3	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
1.0	0.750	22	22	1.0	22	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
1.0	0.875	49	49	1.0	49	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					
1.0	1.000	67	67	1.0	67	1
.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.00
0000	1.000000					

=== by_n ===

Grouped by n (problem size) to study scaling / finite-size effects.

Groups: 19 (metrics computed over successful rows with no n-NaN metric values)

-- eval_advantage --

	n	n_total	n_success	success_rate	n_metric	mean	
	median		p25	p75	iqr	std	min ma
x							
12	30	30	1.0	30	0.523924	0	
.480000	0.241561	0.762213	0.520652	0.313736	0.086957	1.	
0							
13	30	30	1.0	30	0.543272	0	
.504408	0.279121	0.747461	0.468340	0.289682	0.126214	1.	
0							
14	30	30	1.0	30	0.548725	0	
.498340	0.300498	0.803991	0.503493	0.295225	0.135593	1.	
0							
15	30	30	1.0	30	0.544822	0	
.449492	0.278481	0.801054	0.522573	0.291462	0.150943	1.	
0							
16	54	54	1.0	54	0.532249	0	
.448598	0.279981	0.817071	0.537089	0.299130	0.052023	1.	
0							
17	54	54	1.0	54	0.544978	0	
.480218	0.291872	0.792417	0.500546	0.287744	0.106383	1.	
0							
18	84	84	1.0	84	0.547902	0	
.501485	0.266382	0.783873	0.517492	0.288541	0.068273	1.	
0							

19	72	72	1.0	72	0.540443	0
.492052	0.281205	0.801351	0.520146	0.292838	0.064935	1.
0						
20	66	66	1.0	66	0.536468	0
.495068	0.270217	0.793025	0.522808	0.289255	0.186441	1.
0						
21	66	66	1.0	66	0.537154	0
.476596	0.270846	0.806952	0.536106	0.293370	0.102041	1.
0						
22	60	60	1.0	60	0.537038	0
.472805	0.282885	0.802782	0.519897	0.302898	-0.250000	1.
0						
23	48	48	1.0	48	0.545930	0
.491288	0.274405	0.788235	0.513830	0.285521	0.182609	1.
0						
24	54	54	1.0	54	0.538575	0
.502058	0.283946	0.790168	0.506222	0.287431	0.170940	1.
0						
25	42	42	1.0	42	0.552566	0
.500000	0.302811	0.782777	0.479966	0.278518	0.180617	1.
0						
26	36	36	1.0	36	0.550149	0
.434257	0.299945	0.809913	0.509969	0.286200	0.179916	1.
0						
27	30	30	1.0	30	0.560385	0
.522520	0.300757	0.808811	0.508055	0.276094	0.242991	1.
0						
28	24	24	1.0	24	0.531492	0
.474419	0.298564	0.779762	0.481197	0.313430	-0.264957	1.
0						
29	24	24	1.0	24	0.550528	0
.500542	0.270262	0.817143	0.546880	0.284605	0.208511	1.
0						
30	12	12	1.0	12	0.559531	0
.533652	0.292618	0.775901	0.483283	0.280172	0.242553	1.
0						

-- eval_correlation --

n	n_total	n_success	success_rate	n_metric	mean	
median	p25	p75	iqr	std	min	max
12	30	30	1.0	25	0.773333	1
.000000	0.666667	1.0	0.333333	0.315054	0.166667	1.0
13	30	30	1.0	25	0.790769	1
.000000	0.846154	1.0	0.153846	0.349358	0.076923	1.0
14	30	30	1.0	25	0.834286	1
.000000	0.857143	1.0	0.142857	0.287731	0.000000	1.0
15	30	30	1.0	25	0.813333	1
.000000	0.733333	1.0	0.266667	0.288033	0.066667	1.0
16	54	54	1.0	45	0.769444	1
.000000	0.625000	1.0	0.375000	0.338105	0.000000	1.0
17	54	54	1.0	45	0.814379	1
.000000	0.764706	1.0	0.235294	0.291959	0.058824	1.0
18	84	84	1.0	70	0.796825	1
.000000	0.777778	1.0	0.222222	0.338907	0.000000	1.0
19	72	72	1.0	60	0.810526	1
.000000	0.657895	1.0	0.342105	0.289283	0.052632	1.0
20	66	66	1.0	55	0.816364	1
.000000	0.700000	1.0	0.300000	0.257311	0.000000	1.0
21	66	66	1.0	55	0.785281	0
.904762	0.714286	1.0	0.285714	0.311164	0.047619	1.0
22	60	60	1.0	50	0.796364	1
.000000	0.727273	1.0	0.272727	0.305435	0.000000	1.0
23	48	48	1.0	40	0.806522	1
.000000	0.717391	1.0	0.282609	0.312899	0.043478	1.0

24	54	54	1.0	45	0.801852	1
.000000	0.666667	1.0	0.333333	0.301663	0.083333	1.0
25	42	42	1.0	35	0.828571	1
.000000	0.760000	1.0	0.240000	0.274496	0.120000	1.0
26	36	36	1.0	30	0.776923	1
.000000	0.596154	1.0	0.403846	0.339443	0.076923	1.0
27	30	30	1.0	25	0.837037	1
.000000	0.703704	1.0	0.296296	0.240978	0.259259	1.0
28	24	24	1.0	20	0.771429	0
.964286	0.642857	1.0	0.357143	0.317414	0.000000	1.0
29	24	24	1.0	20	0.834483	1
.000000	0.793103	1.0	0.206897	0.278031	0.172414	1.0
30	12	12	1.0	10	0.793333	0
.933333	0.716667	1.0	0.283333	0.297230	0.200000	1.0

-- eval_hamming_frac --

n	n_total	n_success	success_rate	n_metric	mean	
median	p25	p75	iqr	std	min	max
12	30	30	1.0	25	0.113333	0
.000000	0.0	0.166667	0.166667	0.157527	0.0	0.416667
13	30	30	1.0	25	0.104615	0
.000000	0.0	0.076923	0.076923	0.174679	0.0	0.461538
14	30	30	1.0	25	0.082857	0
.000000	0.0	0.071429	0.071429	0.143865	0.0	0.500000
15	30	30	1.0	25	0.093333	0
.000000	0.0	0.133333	0.133333	0.144016	0.0	0.466667
16	54	54	1.0	45	0.115278	0
.000000	0.0	0.187500	0.187500	0.169053	0.0	0.500000
17	54	54	1.0	45	0.092810	0
.000000	0.0	0.117647	0.117647	0.145979	0.0	0.470588
18	84	84	1.0	70	0.101587	0
.000000	0.0	0.111111	0.111111	0.169454	0.0	0.500000
19	72	72	1.0	60	0.094737	0
.000000	0.0	0.171053	0.171053	0.144642	0.0	0.473684
20	66	66	1.0	55	0.091818	0
.000000	0.0	0.150000	0.150000	0.128655	0.0	0.500000
21	66	66	1.0	55	0.107359	0
.047619	0.0	0.142857	0.142857	0.155582	0.0	0.476190
22	60	60	1.0	50	0.101818	0
.000000	0.0	0.136364	0.136364	0.152718	0.0	0.500000
23	48	48	1.0	40	0.096739	0
.000000	0.0	0.141304	0.141304	0.156450	0.0	0.478261
24	54	54	1.0	45	0.099074	0
.000000	0.0	0.166667	0.166667	0.150831	0.0	0.458333
25	42	42	1.0	35	0.085714	0
.000000	0.0	0.120000	0.120000	0.137248	0.0	0.440000
26	36	36	1.0	30	0.111538	0
.000000	0.0	0.201923	0.201923	0.169721	0.0	0.461538
27	30	30	1.0	25	0.081481	0
.000000	0.0	0.148148	0.148148	0.120489	0.0	0.370370
28	24	24	1.0	20	0.114286	0
.017857	0.0	0.178571	0.178571	0.158707	0.0	0.500000
29	24	24	1.0	20	0.082759	0
.000000	0.0	0.103448	0.103448	0.139016	0.0	0.413793
30	12	12	1.0	10	0.103333	0
.033333	0.0	0.141667	0.141667	0.148615	0.0	0.400000

-- eval_bit_accuracy_est --

n	n_total	n_success	success_rate	n_metric	mean	
median	p25	p75	iqr	std	min	max
12	30	30	1.0	30	0.761962	0
.740000	0.620780	0.881107	0.260326	0.156868	0.543478	1.0
13	30	30	1.0	30	0.771636	0
.752204	0.639560	0.873731	0.234170	0.144841	0.563107	1.0

14	30	30	1.0	30	0.774362	0
.749170	0.650249	0.901996	0.251747	0.147613	0.567797	1.0
15	30	30	1.0	30	0.772411	0
.724746	0.639241	0.900527	0.261287	0.145731	0.575472	1.0
16	54	54	1.0	54	0.766124	0
.724299	0.639991	0.908535	0.268545	0.149565	0.526012	1.0
17	54	54	1.0	54	0.772489	0
.740109	0.645936	0.896209	0.250273	0.143872	0.553191	1.0
18	84	84	1.0	84	0.773951	0
.750743	0.633191	0.891937	0.258746	0.144271	0.534137	1.0
19	72	72	1.0	72	0.770222	0
.746026	0.640603	0.900676	0.260073	0.146419	0.532468	1.0
20	66	66	1.0	66	0.768234	0
.747534	0.635108	0.896513	0.261404	0.144627	0.593220	1.0
21	66	66	1.0	66	0.768577	0
.738298	0.635423	0.903476	0.268053	0.146685	0.551020	1.0
22	60	60	1.0	60	0.768519	0
.736402	0.641442	0.901391	0.259949	0.151449	0.375000	1.0
23	48	48	1.0	48	0.772965	0
.745644	0.637202	0.894118	0.256915	0.142760	0.591304	1.0
24	54	54	1.0	54	0.769287	0
.751029	0.641973	0.895084	0.253111	0.143716	0.585470	1.0
25	42	42	1.0	42	0.776283	0
.750000	0.651405	0.891388	0.239983	0.139259	0.590308	1.0
26	36	36	1.0	36	0.775075	0
.717128	0.649972	0.904957	0.254984	0.143100	0.589958	1.0
27	30	30	1.0	30	0.780192	0
.761260	0.650378	0.904406	0.254027	0.138047	0.621495	1.0
28	24	24	1.0	24	0.765746	0
.737209	0.649282	0.889881	0.240599	0.156715	0.367521	1.0
29	24	24	1.0	24	0.775264	0
.750271	0.635131	0.908571	0.273440	0.142302	0.604255	1.0
30	12	12	1.0	12	0.779766	0
.766826	0.646309	0.887950	0.241641	0.140086	0.621277	1.0

SKIP by_n_rho: too many groups (114 > 50)

=== by_eps ===

Grouped by eps (if present) to quantify its effect.

Groups: 4 (metrics computed over successful rows with non -NaN metric values)

-- eval_advantage --

eps	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min
max					
0.15	174	174	1.0	174	0.546652
0.489172	0.295702	0.804367	0.508665	0.284784	-0.250000
1.0					
0.20	216	216	1.0	216	0.540060
0.468696	0.282379	0.791898	0.509519	0.288033	0.120332
1.0					
0.25	252	252	1.0	252	0.544566
0.490835	0.289731	0.806646	0.516914	0.291417	-0.264957
1.0					
0.30	204	204	1.0	204	0.538253
0.492889	0.273973	0.792615	0.518643	0.290809	0.064935
1.0					

-- eval_correlation --

eps	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min max
0.15	174	174	1.0	145	0.814838
1.0	0.727273	1.0	0.272727	0.281750	0.0 1.0
0.20	216	216	1.0	180	0.796602
1.0	0.700000	1.0	0.300000	0.303115	0.0 1.0

0.25	252	252	1.0	210	0.803536
1.0	0.700926	1.0	0.299074	0.304884	0.0 1.0
0.30	204	204	1.0	170	0.795471
1.0	0.666667	1.0	0.333333	0.312661	0.0 1.0

```
-- eval_hamming_frac --
```

eps	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min max
0.15	174	174	1.0	145	0.092581
0.0	0.0	0.136364	0.136364	0.140875	0.0 0.5
0.20	216	216	1.0	180	0.101699
0.0	0.0	0.150000	0.150000	0.151557	0.0 0.5
0.25	252	252	1.0	210	0.098232
0.0	0.0	0.149537	0.149537	0.152442	0.0 0.5
0.30	204	204	1.0	170	0.102265
0.0	0.0	0.166667	0.166667	0.156330	0.0 0.5

```
-- eval_bit_accuracy_est --
```

eps	n_total	n_success	success_rate	n_metric	mean
median	p25	p75	iqr	std	min m
0.15	174	174	1.0	174	0.773326
0.744586	0.647851	0.902183	0.254333	0.142392	0.375000 1
0.0					
0.20	216	216	1.0	216	0.770030
0.734348	0.641190	0.895949	0.254760	0.144016	0.560166 1
0.0					
0.25	252	252	1.0	252	0.772283
0.745417	0.644866	0.903323	0.258457	0.145708	0.367521 1
0.0					
0.30	204	204	1.0	204	0.769126
0.746445	0.636986	0.896308	0.259321	0.145404	0.532468 1
0.0					

=== by_rho_eps ===
 2D grouping by (rho, eps) (only used if number of groups is manageable).
 Groups: 24 (metrics computed over successful rows with no n-NaN metric values)

```
-- eval_advantage --
```

rho	eps	n_total	n_success	success_rate	n_metric
mean	median	p25	p75	iqr	std
min	max				
0.0	0.15	29	29	1.0	29 0.2
45523	0.257642	0.217391	0.293023	0.075632	0.108186 -0.250
000	0.364103				
0.0	0.20	36	36	1.0	36 0.2
47689	0.251667	0.209820	0.284922	0.075102	0.057794 0.120
332	0.412371				
0.0	0.25	42	42	1.0	42 0.2
32437	0.237596	0.199227	0.299285	0.100058	0.110869 -0.264
957	0.413333				
0.0	0.30	34	34	1.0	34 0.2
43674	0.239697	0.218223	0.282779	0.064555	0.070180 0.064
935	0.396552				
0.2	0.15	29	29	1.0	29 0.2
68535	0.271111	0.224490	0.302326	0.077836	0.060354 0.132
530	0.383333				
0.2	0.20	36	36	1.0	36 0.2
69454	0.272810	0.242484	0.299395	0.056911	0.051307 0.171
548	0.383178				
0.2	0.25	42	42	1.0	42 0.2
68744	0.276471	0.246711	0.300334	0.053623	0.069840 0.106
383	0.397959				

0.2	0.30	34	34	1.0	34	0.2
54252	0.254825	0.222789	0.298249	0.075460	0.058994	0.116
466	0.365079					
0.4	0.15	29	29	1.0	29	0.4
09146	0.430380	0.348837	0.469880	0.121042	0.086500	0.200
000	0.548936					
0.4	0.20	36	36	1.0	36	0.3
47585	0.335681	0.287678	0.411525	0.123846	0.091973	0.162
162	0.518367					
0.4	0.25	42	42	1.0	42	0.3
74349	0.377448	0.330421	0.434194	0.103773	0.083496	0.168
000	0.546667					
0.4	0.30	34	34	1.0	34	0.3
60335	0.359243	0.284814	0.438312	0.153497	0.104549	0.151
515	0.616766					
0.6	0.15	29	29	1.0	29	0.5
59467	0.566667	0.516129	0.606426	0.090297	0.073765	0.395
745	0.673307					
0.6	0.20	36	36	1.0	36	0.5
81032	0.595181	0.541537	0.620923	0.079385	0.071937	0.415
584	0.734043					
0.6	0.25	42	42	1.0	42	0.5
88280	0.595462	0.544234	0.630599	0.086365	0.059084	0.471
698	0.698925					
0.6	0.30	34	34	1.0	34	0.5
77994	0.575002	0.520647	0.626129	0.105482	0.076404	0.410
256	0.764706					
0.8	0.15	29	29	1.0	29	0.7
97242	0.805128	0.784314	0.816667	0.032353	0.033761	0.718
876	0.869565					
0.8	0.20	36	36	1.0	36	0.7
94604	0.792130	0.768147	0.816470	0.048323	0.052101	0.696
000	0.891892					
0.8	0.25	42	42	1.0	42	0.8
03587	0.806840	0.785976	0.825815	0.039839	0.041083	0.706
667	0.906433					
0.8	0.30	34	34	1.0	34	0.7
93260	0.792778	0.761477	0.828922	0.067445	0.040503	0.704
225	0.863014					
1.0	0.15	29	29	1.0	29	1.0
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.000
000	1.000000					
1.0	0.20	36	36	1.0	36	1.0
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.000
000	1.000000					
1.0	0.25	42	42	1.0	42	1.0
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.000
000	1.000000					
1.0	0.30	34	34	1.0	34	1.0
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.000
000	1.000000					

-- eval_correlation --

rho	eps	n_total	n_success	success_rate	n_metric	
mean	median		p25	p75	iqr	std
in	max					
0.0	0.15	29	29	1.0		0
NaN	NaN	NaN	NaN	NaN	NaN	NaN
aN	NaN					
0.0	0.20	36	36	1.0		0
NaN	NaN	NaN	NaN	NaN	NaN	NaN
aN	NaN					
0.0	0.25	42	42	1.0		0
NaN	NaN	NaN	NaN	NaN	NaN	NaN

aN	0.0	0.30	34	NaN	34	1.0	0	
	NaN	NaN		NaN	NaN	NaN	NaN	N
aN	0.2	0.15	29		29	1.0	29	0.3
90906	0.400000	0.172414	0.578947	0.406534	0.254495	0.00000	0.00000	
00	0.913043							
0.2	0.20	36		36	1.0	36	0.3	
50961	0.294118	0.164962	0.548583	0.383621	0.247564	0.00000	0.00000	
00	1.000000							
0.2	0.25	42		42	1.0	42	0.3	
09371	0.274436	0.119048	0.460000	0.340952	0.207440	0.00000	0.00000	
00	0.857143							
0.2	0.30	34		34	1.0	34	0.3	
03393	0.230769	0.136288	0.492941	0.356654	0.242760	0.00000	0.00000	
00	0.826087							
0.4	0.15	29		29	1.0	29	0.7	
46753	0.809524	0.700000	0.909091	0.209091	0.240322	0.1304	0.1304	
35	1.000000							
0.4	0.20	36		36	1.0	36	0.6	
75313	0.762353	0.535088	0.883987	0.348899	0.268090	0.0769	0.0769	
23	1.000000							
0.4	0.25	42		42	1.0	42	0.7	
36141	0.804762	0.675000	0.888889	0.213889	0.255652	0.00000	0.00000	
00	1.000000							
0.4	0.30	34		34	1.0	34	0.7	
31662	0.783626	0.626157	0.980769	0.354612	0.263143	0.0526	0.0526	
32	1.000000							
0.6	0.15	29		29	1.0	29	0.9	
36528	1.000000	0.900000	1.000000	0.100000	0.091867	0.6666	0.6666	
67	1.000000							
0.6	0.20	36		36	1.0	36	0.9	
56739	1.000000	0.922619	1.000000	0.077381	0.076484	0.7142	0.7142	
86	1.000000							
0.6	0.25	42		42	1.0	42	0.9	
72169	1.000000	0.948276	1.000000	0.051724	0.049317	0.8461	0.8461	
54	1.000000							
0.6	0.30	34		34	1.0	34	0.9	
45974	1.000000	0.901190	1.000000	0.098810	0.101595	0.5000	0.5000	
00	1.000000							
0.8	0.15	29		29	1.0	29	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
0.8	0.20	36		36	1.0	36	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
0.8	0.25	42		42	1.0	42	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
0.8	0.30	34		34	1.0	34	0.9	
96324	1.000000	1.000000	1.000000	0.000000	0.021437	0.8750	0.8750	
00	1.000000							
1.0	0.15	29		29	1.0	29	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
1.0	0.20	36		36	1.0	36	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
1.0	0.25	42		42	1.0	42	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							
1.0	0.30	34		34	1.0	34	1.0	
00000	1.000000	1.000000	1.000000	0.000000	0.000000	1.0000	1.0000	
00	1.000000							

```

-- eval_hamming_frac --
rho eps n_total n_success success_rate n_metric
mean median p25 p75 iqr std m
in
0.0 0.15 29 29 1.0 0
NaN NaN NaN NaN N
aN
0.0 0.20 36 36 1.0 0
NaN NaN NaN NaN N
aN
0.0 0.25 42 42 1.0 0
NaN NaN NaN NaN N
aN
0.0 0.30 34 34 1.0 0
NaN NaN NaN NaN N
aN
0.2 0.15 29 29 1.0 29 0.3
04547 0.300000 0.210526 0.413793 0.203267 0.127248 0.0434
78 0.500000
0.2 0.20 36 36 1.0 36 0.3
24520 0.352941 0.225709 0.417519 0.191811 0.123782 0.0000
00 0.500000
0.2 0.25 42 42 1.0 42 0.3
45315 0.362782 0.270000 0.440476 0.170476 0.103720 0.0714
29 0.500000
0.2 0.30 34 34 1.0 34 0.3
48303 0.384615 0.253529 0.431856 0.178327 0.121380 0.0869
57 0.500000
0.4 0.15 29 29 1.0 29 0.1
26623 0.095238 0.045455 0.150000 0.104545 0.120161 0.0000
00 0.434783
0.4 0.20 36 36 1.0 36 0.1
62344 0.118824 0.058007 0.232456 0.174450 0.134045 0.0000
00 0.461538
0.4 0.25 42 42 1.0 42 0.1
31929 0.097619 0.055556 0.162500 0.106944 0.127826 0.0000
00 0.500000
0.4 0.30 34 34 1.0 34 0.1
34169 0.108187 0.009615 0.186921 0.177306 0.131572 0.0000
00 0.473684
0.6 0.15 29 29 1.0 29 0.0
31736 0.000000 0.000000 0.050000 0.050000 0.045934 0.0000
00 0.166667
0.6 0.20 36 36 1.0 36 0.0
21631 0.000000 0.000000 0.038690 0.038690 0.038242 0.0000
00 0.142857
0.6 0.25 42 42 1.0 42 0.0
13915 0.000000 0.000000 0.025862 0.025862 0.024658 0.0000
00 0.076923
0.6 0.30 34 34 1.0 34 0.0
27013 0.000000 0.000000 0.049405 0.049405 0.050798 0.0000
00 0.250000
0.8 0.15 29 29 1.0 29 0.0
00000 0.000000 0.000000 0.000000 0.000000 0.000000 0.0000
00 0.000000
0.8 0.20 36 36 1.0 36 0.0
00000 0.000000 0.000000 0.000000 0.000000 0.000000 0.0000
00 0.000000
0.8 0.25 42 42 1.0 42 0.0
00000 0.000000 0.000000 0.000000 0.000000 0.000000 0.0000
00 0.000000
0.8 0.30 34 34 1.0 34 0.0
01838 0.000000 0.000000 0.000000 0.000000 0.010719 0.0000

```

```

00 0.062500
1.0 0.15      29      29      1.0      29 0.0
000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
00 0.000000
1.0 0.20      36      36      1.0      36 0.0
000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
00 0.000000
1.0 0.25      42      42      1.0      42 0.0
000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
00 0.000000
1.0 0.30      34      34      1.0      34 0.0
000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
00 0.000000

```

-- eval_bit_accuracy_est --

	rho	eps	n_total	n_success	success_rate	n_metric	
	mean	median		p25	p75	iqr	std
in	max						m
0.0	0.15	29	29	1.0	29	0.6	
22762	0.628821	0.608696	0.646512	0.037816	0.054093	0.3750	
00	0.682051						
0.0	0.20	36	36	1.0	36	0.6	
23844	0.625833	0.604910	0.642461	0.037551	0.028897	0.5601	
66	0.706186						
0.0	0.25	42	42	1.0	42	0.6	
16218	0.618798	0.599613	0.649643	0.050029	0.055435	0.3675	
21	0.706667						
0.0	0.30	34	34	1.0	34	0.6	
21837	0.619848	0.609112	0.641389	0.032278	0.035090	0.5324	
68	0.698276						
0.2	0.15	29	29	1.0	29	0.6	
34267	0.635556	0.612245	0.651163	0.038918	0.030177	0.5662	
65	0.691667						
0.2	0.20	36	36	1.0	36	0.6	
34727	0.636405	0.621242	0.649698	0.028456	0.025653	0.5857	
74	0.691589						
0.2	0.25	42	42	1.0	42	0.6	
34372	0.638235	0.623355	0.650167	0.026812	0.034920	0.5531	
91	0.698980						
0.2	0.30	34	34	1.0	34	0.6	
27126	0.627413	0.611394	0.649125	0.037730	0.029497	0.5582	
33	0.682540						
0.4	0.15	29	29	1.0	29	0.7	
04573	0.715190	0.674419	0.734940	0.060521	0.043250	0.6000	
00	0.774468						
0.4	0.20	36	36	1.0	36	0.6	
73792	0.667840	0.643839	0.705762	0.061923	0.045987	0.5810	
81	0.759184						
0.4	0.25	42	42	1.0	42	0.6	
87174	0.688724	0.665211	0.717097	0.051886	0.041748	0.5840	
00	0.773333						
0.4	0.30	34	34	1.0	34	0.6	
80167	0.679622	0.642407	0.719156	0.076749	0.052274	0.5757	
58	0.808383						
0.6	0.15	29	29	1.0	29	0.7	
79733	0.783333	0.758065	0.803213	0.045148	0.036882	0.6978	
72	0.836653						
0.6	0.20	36	36	1.0	36	0.7	
90516	0.797590	0.770769	0.810461	0.039693	0.035969	0.7077	
92	0.867021						
0.6	0.25	42	42	1.0	42	0.7	
94140	0.797731	0.772117	0.815300	0.043183	0.029542	0.7358	
49	0.849462						
0.6	0.30	34	34	1.0	34	0.7	

```

88997 0.787501 0.760324 0.813065 0.052741 0.038202 0.7051
28 0.882353
0.8 0.15 29 29 1.0 29 0.8
98621 0.902564 0.892157 0.908333 0.016176 0.016880 0.8594
38 0.934783
0.8 0.20 36 36 1.0 36 0.8
97302 0.896065 0.884074 0.908235 0.024161 0.026050 0.8480
00 0.945946
0.8 0.25 42 42 1.0 42 0.9
01793 0.903420 0.892988 0.912908 0.019920 0.020541 0.8533
33 0.953216
0.8 0.30 34 34 1.0 34 0.8
96630 0.896389 0.880738 0.914461 0.033722 0.020251 0.8521
13 0.931507
1.0 0.15 29 29 1.0 29 1.0
00000 1.000000 1.000000 1.000000 0.000000 0.000000 1.0000
00 1.000000
1.0 0.20 36 36 1.0 36 1.0
00000 1.000000 1.000000 1.000000 0.000000 0.000000 1.0000
00 1.000000
1.0 0.25 42 42 1.0 42 1.0
00000 1.000000 1.000000 1.000000 0.000000 0.000000 1.0000
00 1.000000
1.0 0.30 34 34 1.0 34 1.0
00000 1.000000 1.000000 1.000000 0.000000 0.000000 1.0000
00 1.000000

```

Wrote tidy summary CSV: /Users/julian/CodingProjects/Unit3/QuarticSpeedupK-XOR/logs/eval_stats.csv
Wrote full log: /Users/julian/CodingProjects/Unit3/QuarticSpeedupK-XOR/logs/eval_stats.log

(notebook) Tip: open logs/eval_stats.csv for the full tidy table.